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6 January 1984
SUPERSEDING
MIL-C-21881E
15 February 1977

MILITARY SPECIFICATION

CLOTH, POPLIN, POLYESTER AND COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for polyester and cotton blend poplin fabrics.

* 1.2 Classification. The polyester and cotton fabrics shall be of the following types and classes (see 6.2):

Type II - 4.1 to 4.6 oz.

Class 1 - Khaki 3729
Class 2 - White 3013

Type III - 4.7 to 5.3 oz. (Water Repellent Treated)

Class 1 - Khaki 3726
Class 2 - Green 3452

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the soliciation form a part of this specification to the extent specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-covered and Rayon-covered

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580).

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials. (see 6.6)

3.3.1 Fibers.

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3.3.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate (see 4.4.1).

3.3.1.2 Cotton fiber. The cotton shall be combed (see 4.4.1).

3.3.2 Yarns. The yarns shall be a blend of polyester and cotton drawn and spun into singles yarn for both the warp and the filling when tested as specified in 4.5.

3.3.3 Fiber content. The finished cloth shall contain 65 percent, plus or minus 5 percent, polyester fiber and not less than 30 percent cotton based on the dry weight of the desized specimens when tested as specified in 4.5.

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical requirements

Type	Weight oz/yd^2 (g/m^2) Min.	Max.	Yarns per inch (cm) Minimum Warp	Filling	Breaking Strength lbs (newton) (min) Warp	Filling	Weave
II	4.1 (139.0)	4.6 (155.9)	102 (40.2)	50 (19.7)	120 (534)	50 (223)	Plain
III	4.7 (159.3)	5.3 (179.6)	105 (41.3)	55 (21.6)	150 (667)	70 (311)	Plain

3.4.1 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum width, inclusive of selvage except when woven on shuttless looms the width shall be exclusive of selvages.

3.5 Color

3.5.1 Type II, Class 1. The finished cloth shall be Khaki 3729 and shall match the standard sample (see 6.3).

3.5.2 Type II, Class 2. The finished cloth shall be White 3013 and shall match the standard sample. The cloth shall be fully bleached and maybe supplemented with fluorescent optical brightener (see 4.4.1). The white finished cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.5. When no standard sample is available, the discoloration shall be not less than "fair" when tested as specified in 4.5 (see 6.3).

3.5.3 Type III, Class 1. The finished cloth shall be Khaki 3726 and shall match the standard sample (see 6.3).

3.5.4 Type III, Class 2. The finished cloth shall be Green 3452 and shall match the standard sample (see 6.3).

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3.5.5 Sulfur content. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.5.6 Matching. The color of the finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7000 (± 500) Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 (± 100) Kelvin.

3.5.7 Colorfastness. The type II, class 1, finished cloth shall show fastness to light, laundering (after three cycles), perspiration and crocking equal to or better than the standard sample. When no standard sample is available, the cloth shall show "good" fastness to light, laundering (after three cycles), perspiration and a Munsell value for crocking not lower than 8.5. The type III cloth shall have the same colorfastness requirements, except that combined laundering and bleaching shall be substituted for the three laundering cycles. Tests shall be performed as specified in 4.5. When no standard sample is available, the cloth shall show "good" fastness to combined laundering and bleaching.

3.6 Nonfibrous materials. Prior to application of the water repellent treatment, the starch and protein content, including chloroform-soluble and water-soluble material, shall not exceed 2.0 percent when tested as specified in 4.5.

3.7 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.0 when tested as specified in 4.5.

3.8 Dimensional stability. The finished cloth shall not shrink nor elongate more than 2.0 percent in either the warp or the filling when tested as specified in 4.5. The pre-shrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.9 Seam efficiency. The finished cloths shall have a seam efficiency of not less than 80 percent when tested as specified in 4.5.

3.10 Finish. The cloth shall be desized, scoured, heat set and dyed or bleached (as applicable) to meet the requirements of the standard sample. The type III cloths shall be given an approved durable water repellent treatment (see 6.4) to meet the requirements of 3.10.1 and 3.10.2.

3.10.1 Spray rating. The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than ratings 90, 90, 80 when tested as specified in 4.5.

3.10.2 Dynamic absorption. The initial dynamic absorption shall be an average of 25 percent, maximum. The dynamic absorption after three launderings shall be an average of 35 percent, maximum. Tests shall be performed as specified in 4.5.

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3.11 Length and put-up. Unless otherwise specified (see 5.1.1 and 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.56m). Each length shall be put-up in full width rolls as specified in PPP-P-1134 with the face inward.

3.12 Marking. The face of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.13 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$) when calculated as specified in Section 4, shall not exceed the established point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (See 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection. When required, the first article of finished cloth shall be visually examined for color, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be visually examined and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the contractor shall furnish a certificate for compliance for the polyester fiber identification, the cotton fiber identification, the combing of the cotton fibers and the bleaching and fluorescent whitener for the Type II, class 2 cloth.

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4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards, (91.4m) the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. All defects as defined in Section III of FED-STD-4, which are clearly visible at normal inspection distance (3 feet) (0.91m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.5), "D" for slubs and "C" for knots, shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceed 38.0 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls in the second sample of 20 exceed the individual point value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size X 3600}}{\text{Contracted width of cloth (inches) X Total yards inspected}} = \text{Points per 100 sq yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.62cm) in any dimension - one point
For defects exceeding 3 inches (7.62cm) but not exceeding 6 inches (15.24cm) in any dimension - two points
For defects exceeding 6 inches (15.24cm) but not exceeding 9 inches (22.86cm) in any dimension - three points
For defects exceeding 9 inches (22.86cm) in any dimension - four points

NOTE: The following defects, when present, shall be scored 4 points for each yard (0.91m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
Excessive neppiness

4.4.2.2 Examination for length

4.4.2.2.1 Individual rolls. During the yard-by-yard examinations, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards (1.82m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

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4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, identification of pre-shrinking process and non-conformance with the Textile Fiber Products identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinking process by name or trademark on the cloth or ticket, or two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.2.5 Examination for shade match and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.5.6) and finish (see 3.10). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish. The sample unit shall be a 4 inch by 20 inch (10.2 by 50.8cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.4.3 Examination of Packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing final results. The sample unit for test purposes of Type II cloths shall be 4 continuous yards (3.64m), full width of the finished cloth. The sample unit for test purposes of Type III cloths shall be 5 continuous yards (4.55m) full width of the finished cloth and 1/4 yard (0.22m) full width of the cloth prior to water repellent treatment for determination of starch and protein content. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size yards (.91meter)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test methods

Characteristic	Requirement Paragraph	Test Method
Fiber content		
Polyester (percent)	3.3.3	2100, 4.5.1
Cotton (percent)	3.3.3	2100, 4.5.1
Yarn (singles)		
Warp	3.3.2	Visual 1/
Filling	3.3.2	Visual 1/
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
Discoloration	3.5.2	5660 1/ 2/
Sulfur content	3.5.5	2020
Weave	3.4	Visual 1/
Colorfastness to:		
Light	3.5.7	5660
Laundering (after 3 cycles)	3.5.7	5610 5/ 6/
Perspiration	3.5.7	5680
Crocking	3.5.7	5651
Laundering and bleaching	3.5.7	5605 6/
Nonfibrous materials	3.6	2611
pH	3.7	2811
Dimensional stability	3.8	5556 3/
Seam efficiency	3.9	5110 4/
Water resistance		
Spray rating	3.10.1	5526
Dynamic absorption		
Initial	3.10.2	5500
After 3 launderings	3.10.2	5500 & 5518

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1/ One determination per sample unit and the result reported as "pass" or "fail".
2/ Method 5660 shall be used except that exposure of the specimen standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "satisfactory". Any specimen showing discoloration greater than that of the standard sample shall be rated "unsatisfactory". When no standard sample is available the discoloration of the specimen shall be rated as "excellent", "good", "fair", or "poor", as defined in method 5660.

3/ Cotton laundering procedure shall be followed

4/ The needle size shall be 0.036 (± 0.001) inch (0.91 ± 0.025mm) across the blade eye. The thread shall be polyester covered conforming to MIL-T-43548, Ticket No. 50, 3 ply for the needle and Ticket No. 70, 2 ply for the looper. As an alternate, a cotton thread conforming to Type 1A1 of V-T-276 may be used. Ticket No. 60, 3 ply for the needle and Ticket No. 70, 3 ply for the looper.

5/ The specimens must be dried after each of the 3 laundering cycles.
6/ Only the stain on the cotton and polyester fibers of the color transfer cloth shall be evaluated.

4.5.1 Polyester and cotton fiber content. The general procedures of method 2100 shall be followed with the fiber content percentages calculated as follows:

$$\text{Polyester, percent} = \frac{\text{Weight of dry residual fiber X 100}}{\text{Weight of dry desized specimen}}$$

Percent cotton = 100 - percent polyester

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

5. PACKAGING

5.1 Preservation-Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. The cloth put up as specified (see 3.11) shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use. Type II cloths covered by this specification are intended for use in shirts worn by Naval personnel. Type III cloths are intended for use by all services in swimming and general utility trunks for male personnel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Type and class required (see 1.2)
c. When first article sample is required. The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.
d. Width required (see 3.4.1)
e. Minimum and maximum lengths for cloth (see 3.11)
f. Levels of packaging and packing required (see 5.1 and 5.2)

6.3 Standard sample. For access to the standard sample, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Durable water repellent. Durable water repellent Treatment (see 3.10) for use of type III cloth is the responsibility of the U.S. Army Natick Research and Development Center, Natick, MA 01760, and is based on extensive tests including those for toxity, which are not set forth in this specification. Because of the time necessary to conduct full evaluation only those treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for related procurement. Information on currently approved material may be obtained from Defense Personnel Support Center, Philadelphia, PA 19101.

* 6.5 Fabric defect scales. Fabric Defect Replica Kits are available from Sears, Roebuck and Company, Department 817 (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy - NU
Air Force - 99

Review activities:
Army - MD
Air Force - 82
DLA - CT

User activities:
Navy - CG, MC

Preparing activity:
Navy - NU

Project No. 8305-0834

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